

**Dynamics of host resistance against
Rhizoctonia bataticola (Taubenh.) Butler
causing dry root rot in Chickpea (*Cicer arietinum* L.)**

DISSERTATION REPORT

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In

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By

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2007



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No part of this thesis has been submitted for any other degrees in other university or institution. All the assistance and help received during the course of investigation have been duly acknowledged by her.

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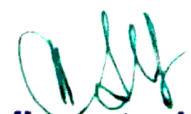
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I hereby declare that the dissertation entitled "**Dynamics of host resistance against *Rhizoctonia bataticola* (Taubenh.) Butler causing dry root rot in chickpea (*Cicer arietinum* L.)**" is submitted for the partial fulfillment of the degree of "**Master of Science in Microbiology**" at the Department of Plant Breeding and Genetics" JNKVV, Jabalpur (M.P.) is an authentic record of my work carried out under the supervision of **Dr. (Smt.) Om Gupta**, Principal Scientist, Plant Pathology & Project Incharge AICRP on Chickpea, Department of Plant Breeding & Genetics, Jawaharlal Nehru Krishi Vishwa Vidyalaya, Jabalpur (M.P.).

The matter embodied in this dissertation is original and not substantially the same as one which has been submitted for a degree, diploma or any other academic qualification at any other university in India/abroad.

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(Manisha Rathi)

CONTENTS

S. No.	Title	Page No.
1.	Introduction	1-3
2.	Review of Literature	4-9
3.	Materials and Methods	10-15
4.	Results	16-28
5.	Discussion	29-31
6.	Summary, Conclusion and Suggestions	23-33
7.	Bibliography	34-37

LIST OF TABLES

Table No.	Title	Page No.
	Reaction of chickpea entries/genotypes against dry root rot through Blotter paper technique	
1.	Donors	21
2.	Mutant	21
3.	ICRISAT chickpea wilt, root rot nursery (CWRRN)	22
4.	National nursery-Dry root rot (DRRN)	23
5.	Promising entries from National nursery wilt (NNW)	24
6.	Station trial-Desi (S.T.D.)	25
7.	Promising lines/Breeding material	26
8.	Germplasm-(NBPGR)	27
9.	List of entries/genotypes scored as rating '1' against dry root rot <10 per cent	28

LIST OF PLATES

Plates No.	Title	In between page No.
1.	Plant showing symptoms of dry root rot affected by <i>Rhizoctonia bataticola</i> A) Dry root rot affected chickpea plant Growth pattern and pigmentation of isolate of <i>Rhizoctonia bataticola</i> on Potato dextrose agar medium B) Front view C) Dorsal view Morphological characters of <i>Rhizoctonia bataticola</i> as observed in microscope D) Septation of mycelia E) Formation of sclerotia	16-17 16-17 16-17
2.	F) Growth of <i>Rhizoctonia bataticola</i> on Potato dextrose broth G) Trays showing seedlings of test line inoculated with <i>Rhizoctonia bataticola</i> in an environmental test chamber for 8 days at 35°C	17-18
3.	Disease reaction of isolate of <i>Rhizoctonia bataticola</i> on chickpea entries/genotypes alongwith susceptible check BG 212 and control (rating scale : 1-9) H) Susceptible check : BG 212 I) Control : Uninoculated seedlings J) Resistant (R) : 1 K) Moderately Resistant (MR) : 2 and 3	20-21
4.	L) Tolerant (T) : 4 and 5 M) Moderately Susceptible (MS) : 6 and 7 N) Susceptible (S) : 8 and 9	20-21

ABBREVIATIONS

%	per cent
@	at the rate of
μm	micro meter
$^{\circ}\text{C}$	degree centigrade
cm	centimeter
<i>et al.</i>	co workers
g	gram
ha	hectare
ha^{-1}	per hectare
hrs.	hours
i.e.	that is
mg	milligram
ml	milliliter
mm	millimeter
kg	kilogram
N	nitrogen
psi	per sequare inch
<	less than
>	greater than
viz.,	namely
ICRISAT	International Crop Research Institute for Semi Arid Tropics
NBPGR	National Bureau of Plant Genetic Resources



Chapter I

Introduction

INTRODUCTION

Chickpea (*Cicer arietinum* L.) is commonly known as Bengal gram (Indian), Chickpea (English), Garbanzo (Latin America), Hommes (Arab World), Nohud, Lablabi (Turkey), Shimbra (Ethiopia). It is the third most important food legume globally grown in over 40 countries and first grown in Turkey about 7,000 B.C. There are three chickpea types, desi type is grown in semi arid zones of India and Middle eastern countries while kabuli type is grown in temperate region and gulabi type is restricted to only certain districts of Madhya Pradesh (Muehlbauer and Singh, 1987; Malhotra *et al.*, 1987). The most important chickpea producing countries are India (64 per cent), Turkey (8 per cent), Pakistan (7 per cent), Iran (3 per cent), Mexico (3 per cent), Myanmar (3 per cent), Ethiopia (2 per cent), Australia (2 per cent) and Canada (1 per cent).

Chickpea has the highest nutritional composition and does not contain any specific major anti-nutritional factors. On an average, chickpea seed contains 23 per cent protein, 38 to 59 per cent total carbohydrates, 5 per cent fat, 6 per cent crude fiber, 6 per cent soluble sugar and 3 per cent ash. The mineral component is high in phosphorous, calcium, magnesium, iron and zinc which were 340, 190, 140, 7 and 3 mg per 100 g respectively.

Chickpea is mainly used for human consumption and only a small proportion is used as feed. The kabuli type (white or cream seed coat) is generally used as whole grains, while desi type (colored seed coat) is used as whole seed, dehulled splits or

flour. Chickpea is also known for its use in herbal medicine and cosmetics.

Chickpea meets 80 per cent of its nitrogen requirement from symbiotic nitrogen fixation and can fix upto 140 kg N ha^{-1} from air. It leaves substantial amount of residual nitrogen behind for subsequent crops and adds much needed organic matter to maintain and improve soil health, long term fertility and sustainability of ecosystems.

Area, production and productivity

According to National level monitoring team report 2005-06 chickpea occupied an area of 67.148 lakh ha in India having total production of 54.694 lakh tonnes with the productivity of 815 kg ha^{-1} . Whereas it was grown in 27.465 lakh ha in Madhya Pradesh which gave total production of 25.489 lakh tonnes with average yield of 928 kg ha^{-1} (2004-05).

However its productivity is affected by several biotic and abiotic stresses. Among biotic especially soil borne pathogens causing collar rot, wilt and dry root rot are of great significance. Among them *Sclerotium rolfsii* (collar rot), *Fusarium oxysporum* f. sp. *ciceri* (vascular wilt) and *Rhizoctonia bataticola* (dry root rot) are important. Combining useful economic traits with multiple disease resistance and wider adaptability is the demand of the day particularly in Rice fallow and Soybean chickpea based cropping systems.

A very little information is available in literature with regards to dry root rot resistance in chickpea. However, numbers

of wilt resistant varieties from all over the country are now available and are in cultivation.

Therefore, keeping the above facts in view an investigation entitled "**Dynamics of host resistance against *Rhizoctonia bataticola* (Taubenh.) Butler causing dry root rot in chickpea (*Cicer arietinum* L.)**" has been undertaken with the following objectives:

To test the pathogenicity and to identify the sources of host resistance against dry root rot in different genotypes of desi and kabuli chickpea.

Chapter II
Review of Literature

REVIEW OF LITERATURE

In recent years, many improved varieties of chickpea have been evolved, which can be grown under irrigated as well as rainfed situations in different agro climatic conditions. These varieties have high yield potential and tolerance to many diseases. But the harsh environmental conditions under which chickpea are generally grown and the infestation of different diseases particularly dry root rot (*Rhizoctonia bataticola*) impose restrictions on the expressions of its genetic yield potential. Since the new high yielding varieties of gram are of recent introduction, hence the experimental evidences on their resistance against dry root rot are somewhat scanty. However, attempts have been made to review the available literature critically and elucidated here.

2.1 Occurrence and losses

The first report of the occurrence of root rot in chickpea along with wilt was made by Padwick (1948). The incidence of root rot ranging from 3.58 to 20.63 per cent in 30 villages of northern Madhya Pradesh was reported by Gupta *et al.* (1983). This is one of the most common diseases in south east Asian countries and limits chickpea production (FAO, 1990). The disease is serious in areas where the ambient day temperatures are around 30°C at the seedling or at the flowering stage. In India this disease occurs mainly in the central and southern states, where a lot of mortality at the flowering and podding stages is due to this disease.

Tripathi and Sharma (1983) studied the incidence of chickpea dry root rot (*Rhizoctonia bataticola*) in southern Haryana

and reported that the incidence was high from late October to mid November which was described in December and January and then again increased in the months of February and March. They also noted that seed sown at a depth of 15 cm developed a less root rot than that sown at 7.5 cm.

Nene *et al.* (1984) observed that chickpea is attacked by a large number of diseases. Among these, dry root rot caused by *Rhizoctonia bataticola* is the most devastating and is widely prevalent in most of the chickpea growing areas of the world. In India this disease causes heavy losses to chickpea crop.

Ahmed *et al.* (1991) noted that mortality of the susceptible entry ICC 4951 was 100 per cent during the screening. The mean plant mortality ranged from 2.3-7.5 per cent.

Singh *et al.* (2002) conducted a survey in Vindhyan plateau zone of Madhya Pradesh, India to evaluate the prevalence of wilt and dry root rot diseases in chickpea. They reported that *Fusarium oxysporum* was most prevalent (56 per cent) in wilted plants followed by *Rhizoctonia bataticola* (17 per cent) and *Trichoderma* sp. (10 per cent). The incidence of *Rhizoctonia bataticola* also caused withering and drying of plants, ranged from 8 to 20 per cent.

2.2 Screening techniques

The Blotter paper technique (Nene *et al.*, 1981) is routinely used at ICRISAT to confirm resistance in test lines identified promising in the field. In this technique roots of 5 to 7 days old seedlings raised in the sterilized sand are inoculated with inoculum and incubated at 30 to 35°C for 8 days. At the end of

the incubation, seedlings are examined for the extent of root damage and scored on 1 to 9 point rating scale.

To confirm the resistance in the field, growing of resistant entries/genotypes in a wilt sick plot have been reported good for DRR. The only difference is to plant a cultivar that is susceptible to *R. bataticola*, but resistant to *F. oxysporum* f. sp. *ciceri*. Examples of such cultivars are BG 212 and ICC 229 (Nene et al., 1981).

Gupta and Babbar, (2006) conducted a field experiment to search the multiple as well as individual disease resistant sources against various soil borne diseases in chickpea. The results revealed that the pathogens *Sclerotium rolfsii* and *R. bataticola* caused severe mortality ranging from 1-74 and 1-100 per cent at seedling and later stages of crop growth. However, the genotypes, which performed well, exhibited <10 per cent mortality against these diseases.

2.3 Symptoms

Narsimhan (1929) observed that plants infected with *R. bataticola* initially showed drooping of leaves followed by wilting. The main roots of such plants show blackening of finer roots. Reddy et al. (1991) stated that the countries like Mandalay, Sayaing and Bago, central and lower Myanmar the major disease in chickpea was dry root rot caused by *R. bataticola*, Aung et al. (1992) observed that dry root rot caused by *R. bataticola* was major disease of chickpea.

Nema and Khare (1973) reported that chickpea cultivation in Madhya Pradesh was greatly threatened by soil borne diseases generally appearing at two phases of growth, early seedling stage

and flowering stage, wilting was due to *Fusarium oxysporum* f. sp. *ciceri*, root rot caused by *Rhizoctonia bataticola* and *Sclerotium rolfsii* resulted in collar rot.

Khalid and Ilyas (2000) evaluated 108 chickpea germplasm lines for sources of resistance against *Macrophomina phaseolina* infection and disease assessment. On the basis of external expression of dry root rot symptoms, revealed that none of the tested lines was highly resistant, only one line (971120) was resistant and another line (97104) was moderately resistant.

2.4 Taxonomy

The disease pathogen of dry root rot under study belongs to the species *Phaseolina* of genus *Macrophomina* of the family Spharopsidaceae in class Deuteromycetes and sub-class Coelomonycetidae in the order Spharopsidales. It comes under the division Amastigomycota and sub-division Deuteromycotina of the kingdom Mycetae.

2.5 Pathogen

Pathogen *Rhizoctonia bataticola* (Taubenh.) Butler as a Plant Pathologist was recognized by Halsted in (1890) has a host range of over 500 plant species (Jones and Canada, 1994) with world wide distribution and with variable characteristics. On the infected plants sclerotia are seen in the pith region and outer of the roots just below the bark region. Sclerotia are brown to jet black, minute smooth, externally hard, variable in shape: globose, oval, oblong, elliptical or even forked and varying in size (25x22 to 152x32 μm) produced abundantly in the infected host tissues (Subramanian, 1971).

2.6 Host resistance

Ananda Rao and Haware (1987) identified several lines of chickpea resistant to dry root rot at ICRISAT. Some of these lines have the resistance to wilt and dry root rot. The inheritance of resistance to dry root rot of chickpea seems to be monogenic with resistance dominant over susceptibility.

Nene *et al.* (1989) identified broad based stable resistant lines to wilt and root rots on the basis of multilocation evaluation, which are ICC 11550, ICC 9032, ICC 9023, ICC 2862 and ICC 11551.

Reddy *et al.* (1990) reported two chickpea varieties i.e. Avrodhi and ICC 98 resistant to wilt and root rots from ICRISAT during 1985-86 to 1988-89.

Bhatt (1993) observed that BGM 416, BG 416, ICC 1376 and IC 11314 were resistant to root rot. Resistant cultivars had greater number of total roots during early growth phases.

Dermiro *et al.* (1994) observed that dry root rot disease of gram is also becoming a serious disease affecting productivity of this crop. They isolated *Rhizoctonia bataticola* from plants and seeds of chickpea cultivar Azizive 94 and tested pathogenicity.

Gupta (1997) identified genotypes ICCV 89402, ICCV 90201, ICCV 90254, BG 209, H 88-2, GF 88426 and RSG 180 having dual resistance against wilt and dry root rot.

Gangwar *et al.* (2002) screened 35 chickpea cultivars for resistance to dry root rot caused by *Rhizoctonia bataticola* in a field experiment. All the cultivars exhibited disease symptoms, only 10 cultivars (ICC 2644, 10384, 10630, 112244, 11332, ICCL

81002, 810810, ICC 12263, 12441 and ICCV 90254) were resistant whereas 5 genotypes (GCP 9504, Phule G 96020, 96105, 96313 and GL 91059) were moderately resistant to dry root rot.

Prajapati *et al.* (2003) while working on resistant sources of chickpea against dry root rot (*Rhizoctonia bataticola*) evaluated forty-five chickpea cultivars for resistance to dry root rot. Among these, H 355, JG 315, Pusa-212, G 543, ICC 32, BG 244, Phule, G 5, ICC numbers 2644, 10384, 11224, 10630 and 11332, ICCV 10, ICC numbers 12441 and 12263, 81010 and 81002, Avrodhi, ILC 702 and ILC 200 cultivars were rated as resistant. 5 cultivars were moderately resistant and 9 were tolerant.

Panday *et al.* (2004) evaluated twenty-nine chickpea germplasm accessions, 10 cultivars and 8 advanced breeding lines for resistance to dry root rot, caused by *Rhizoctonia bataticola* under *in vitro* conditions. They observed that one germplasm accession (ICC 14395), a cultivar (ICCV₂) and advanced breeding line were resistant to dry root rot. Of the remaining lines 22 were moderately resistant, 19 susceptible and 2 highly susceptible line (BG 212 and ICC 12267) used as the control.

Singh *et al.* (2002) conducted a survey in Vindhyan plateau zone of Madhya Pradesh, India to evaluate the prevalence of wilt and dry root rot disease in chickpea. They reported that *Fusarium oxysporum* was most prevalent (56 per cent) in wilted plants followed by *Rhizoctonia bataticola* (17 per cent) and *Trichoderma* sp. (10 per cent). The incidence of *Rhizoctonia bataticola* also caused withering and drying of plants, which ranged from 8 to 20 per cent.



Chapter III

Materials and Methods

MATERIALS AND METHODS

The present investigation entitled "**Dynamics of host resistance against *Rhizoctonia bataticola* (Taubenh.) Butler causing dry root rot in chickpea**" was undertaken to identify the sources of host resistance for dry root rot (DRR) in different genotypes of desi and kabuli chickpeas and to test the pathogenicity. The investigation was carried out on occurrence of dry root rot caused by *Rhizoctonia bataticola* in different genotypes of chickpea under *in vitro* conditions. The materials used and methods adopted are as follows.

3.1 Experimental site

The experiment was conducted under *in vitro* condition in the chickpea Pathology laboratory under All India Coordinated Research Project (AICRP) on chickpea, Département of Plant Breeding and Genetics, JNKVV, Jabalpur.

3.2 Collection of dry root rot affected samples

The plants sample showing the symptoms of dry root rot were collected from the research experiments conducted at Seed Breeding Farm under All India Coordinated Research Project on chickpea.

3.3 Collection of chickpea genotypes

The seeds of chickpea genotypes used in the present investigation were also obtained from the AICRP on chickpea sub centre, Jabalpur and International Centre for Research in Semi

Arid Tropics (ICRISAT), Hyderabad and National Bureau of Plant Genetic Resources (NBPGR), New Delhi.

3.4 Materials used

Borosil make glasswares were used for isolation, purification and identification throughout the investigation. All the glasswares were cleaned with the solution made from detergent powder and subsequently rinsing with clean water. Petri-plates were sterilized in hot air oven at 180°C for 2 hrs. The metallic instruments were sterilized by dipping them in alcohol after heating over burner flame.

The culture media were sterilized in an autoclave at a pressure of 105 kg/p.s.i. (15 lbs, p.s.i./at 121.6°C) for 15 minutes. At the time of pouring in the Petri-plates streptomycin @ 600 mg/l was added to the medium for inhibiting the bacterial growth. The soil/sand and chickpea straw was sterilized at 2.1 kg pressure/square inch (30 lbs pressure p.s.i.) for 30 minute.

Media

The following media were used during the course of investigation.

Non synthetic media

a) Potato dextrose agar (PDA)

Peeled potato	-	200 g
Dextrose	-	20 g
Agar	-	20 g
Distilled water	-	1000 ml

b) Potato dextrose broth (PDB)

Peeled potato	-	200 g
Dextrose	-	20 g
Distilled water	-	1000 ml

3.5 Isolation, Purification and Identification

Isolation was done from the affected dry root rot plants of chickpea. Small pieces of about 1 cm length of the affected roots were surface sterilized with 1:1000 mercuric chloride solution followed by washing it thrice with sterilized distilled water. The root pieces were dried on sterilized blotting paper before plating on PDA over the Petri-plates kept for incubation at 25°C for further studies.

The culture of *Rhizoctonia bataticola* was purified by using the hyphal tip technique and incubated at 25°C for further growth. The purified culture was maintained on PDA slants.

The purified culture was identified on the basis of morphological characters.

3.6 Pathogenicity test

3.6.1 Pot inoculation technique

The culture was mixed in the sterilized soil/sand mixture (1:1) in the ratio of 1:9 and it was filled in the earthen pots (30 cm diameter) then these pots were watered. Seeds of susceptible variety BG-212 were surface sterilized in 2.5% sodium hypochlorite for 5 minutes. After that 30 seeds of each genotype

were sown in the above pots at 2 cm depth after 4 days of inoculating the culture. The pots were watered adequately and regularly but not excessively to get proper germination of the chickpea seeds. Observations on plant mortality due to dry root rot were recorded and reisolation was done from the roots of affected plants to confirm the pathogenicity.

3.6.2 Screening technique

Blotter paper technique was employed (Nene *et al.*, 1981), for this purpose 5 mm disc of the culture was placed on PDA poured Petri-plates and incubated at 25°C for 4 days. Five mm disc was cut from the above culture and transferred to 250 ml flasks each containing 100 ml of PDB. After 5 days of incubation at 25°C two mycelial mats were removed from the flask which were added to 100 ml sterilized distilled water in a beaker after its proper crushing for 1 minute in the blender.

Surface sterilized seeds of different chickpea genotypes were sown on plastic trays containing sterilized soil + sand (1:1). Five days old seedlings of each genotype were uprooted in such a way so that root system is not disturbed. The root system of these seedlings was properly washed in running water followed by rinsing in sterilized distilled water. The roots of all genotypes (test lines) were dipped in the inoculum kept in a beaker with an up and down movement for about 30 seconds and the excess inoculum was removed by touching the roots to the edge of beaker.

Twenty seedlings of the each test line were taken and ten of these were kept separately on two blotter papers (size 45 cm x 25 cm with one fold). The blotter paper was moistened adequately and the seedlings were kept in such a way so that only cotyledons and roots are covered and the green tops of the seedlings remains out side the blotter paper. The seedlings of a susceptible check (BG 212) were also inoculated with each batch of test seedlings.

The folded blotter papers were kept one on top of the other, in heaps of ten in a tray. A susceptible check was kept in each heap. These trays were placed into the incubator at 35°C for 8 days. Artificial light was provided for 12 hrs and the blotter papers were moistened adequately on alternate day.

3.7 Rating scale

In order to find out the resistant genotype of chickpea, test lines were scored at the end of the incubation period by examining the seedlings for the extent of root damage and were scored for the disease on 1-9 point scale as mentioned below:

Rating scale used for disease scoring

Rating	Category	Symptoms of DRR	DRR%
1	Resistant	No infection on roots	0.0-10
2-3	Moderately resistant	Very few small lesions on roots	10.1-20
4-5	Tolerant	Lesions on roots clear but small, new roots free from infection	20.1-30
6-7	Moderately susceptible	Lesions on roots many, new roots generally free from lesions	30.1-40
8-9	Susceptible	Roots infected and completely discolored	40.1 and above

Test lines showing ratings 1-3 are considered acceptable for the breeding programme, ratings 4-5 are acceptable only if lines with ratings 1-3 are not available, ratings 6-9 are not acceptable.

After recording the observations on disease resistance, percentage of resistant, moderately resistant, tolerant, moderately susceptible and susceptible seedlings was worked out with the help of formula given below:

$$\text{Percent of infected plants} = \frac{\text{Number of infected plants}}{\text{Total number of plants}} \times 100$$



Chapter IV

Results

RESULTS

4.1 Collection of dry root rot infected plants

Dry root rot infected plants of chickpea were collected from multiple disease sick field located at Seed Breeding Farm, Jawaharlal Nehru Krishi Vishwa Vidyalaya, Jabalpur, during February to March 2007.

4.2 Symptoms of dry root rot

Dry root rot in chickpea plants was observed in the field at flowering to podding stage. The leaves and stems of affected plants showed straw-coloured appearance. The disease appeared in a severe form when ambient temperature was between 25 and 30°C. The pathogen forms reddish brown sunken lesions on the stem and tap root most frequently near the soil line causing stunting growth or death of the plant.

The roots were usually dry unless the soil is wet. The tap root is quite brittle, showed shredding of the bark and broken very easily. Minute dark black sclerotial bodies were observed with the aid of a 10 x hand lens on the surface of the root as well as in the pith (Plate 1: A).

4.3 Pathogen

The pathogen isolated from infected plants exhibited the growth of *Rhizoctonia bataticola* on PDA at 25°C (Plate 1: B to E). Its growth pattern in the colony appears to be somewhat cottony, fluffy and light brown to dark brown or greyish black in colour. Observations of the dry root rot of infected plants revealed the

1. Plant showing symptoms of dry root rot affected by *Rhizoctonia bataticola*

A) Dry root rot affected chickpea plant

Growth pattern and pigmentation of isolate of *Rhizoctonia bataticola* on Potato dextrose agar medium

B) Front view

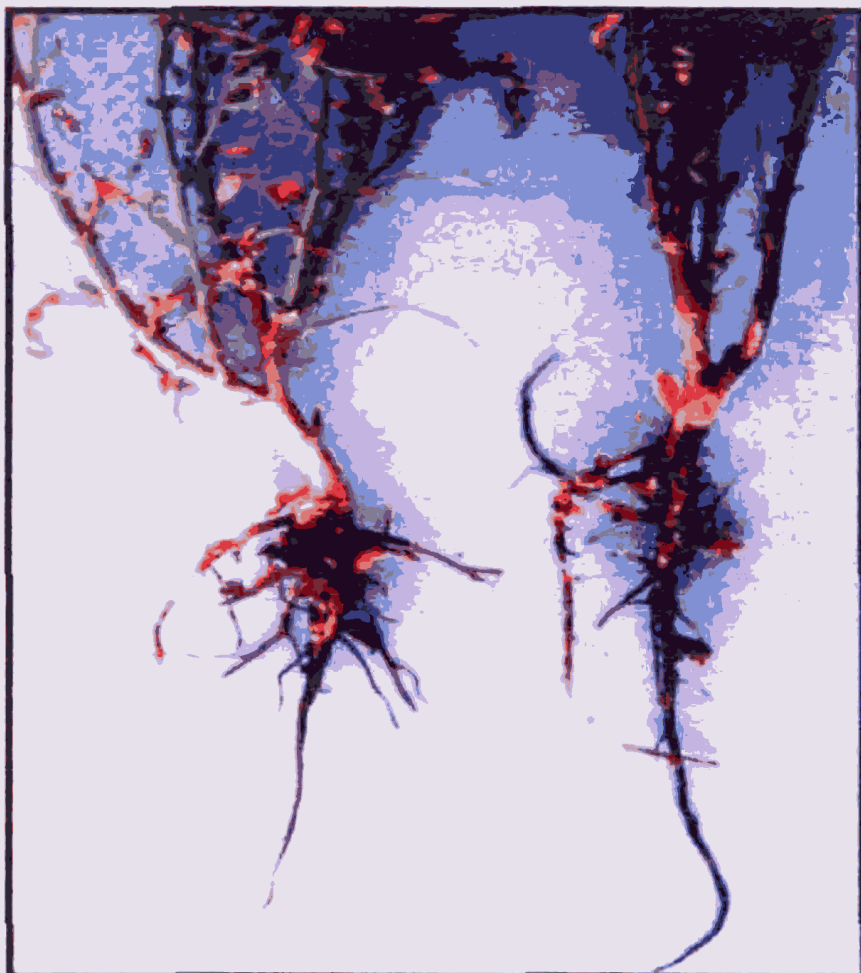
C) Dorsal view

Morphological characters of *Rhizoctonia bataticola* as observed in microscope

D) Septation of mycelia

E) Formation of sclerotia

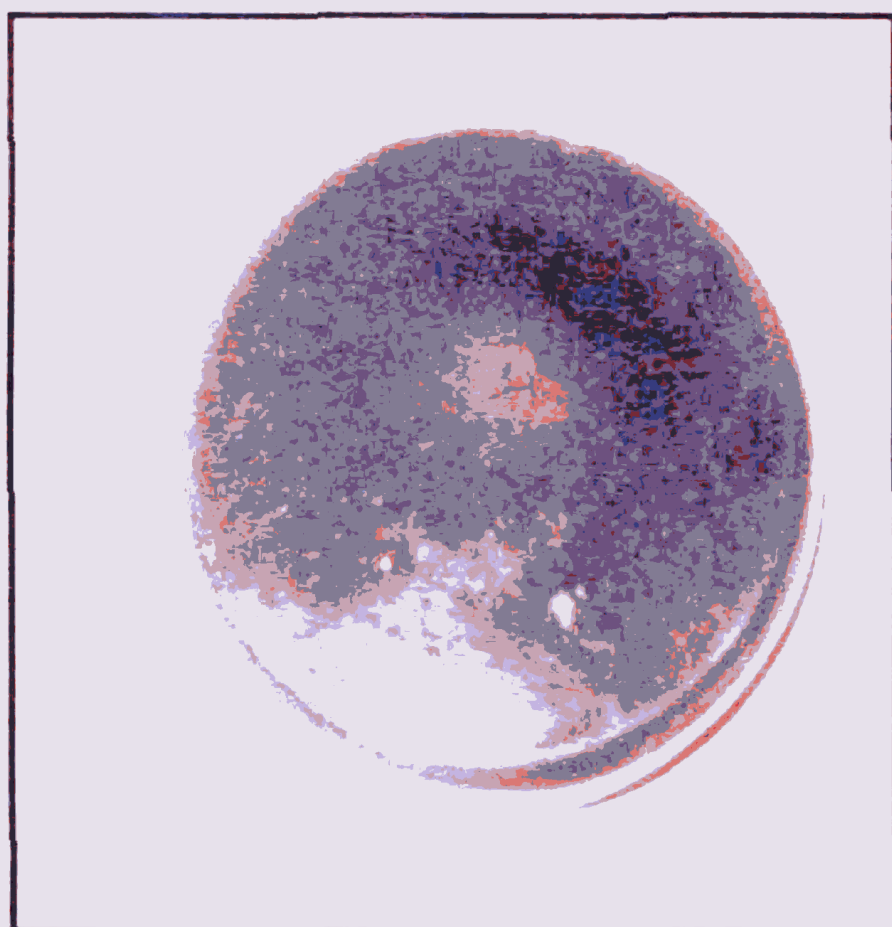
Plate I



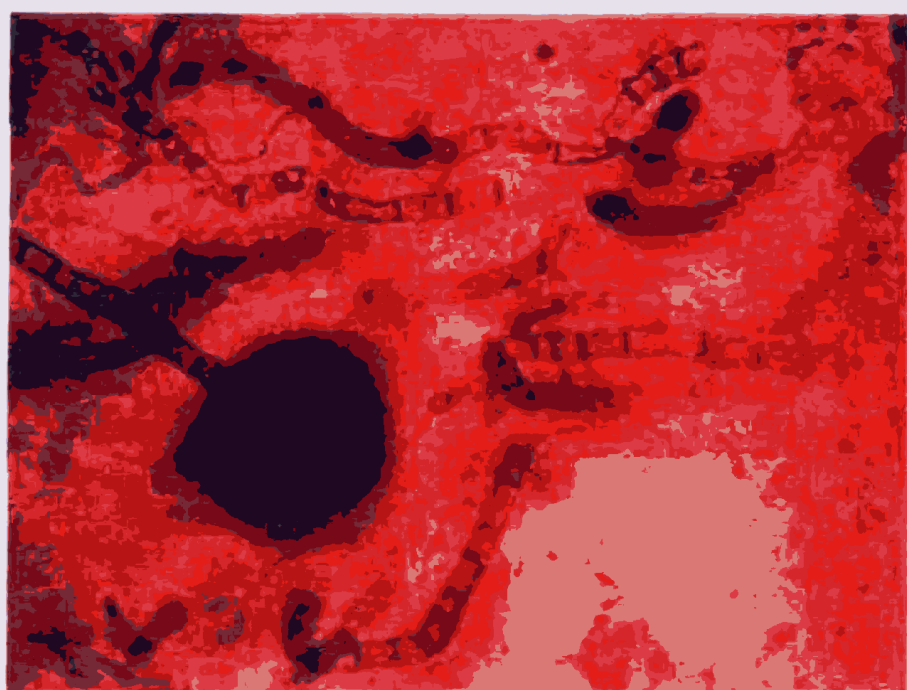
A



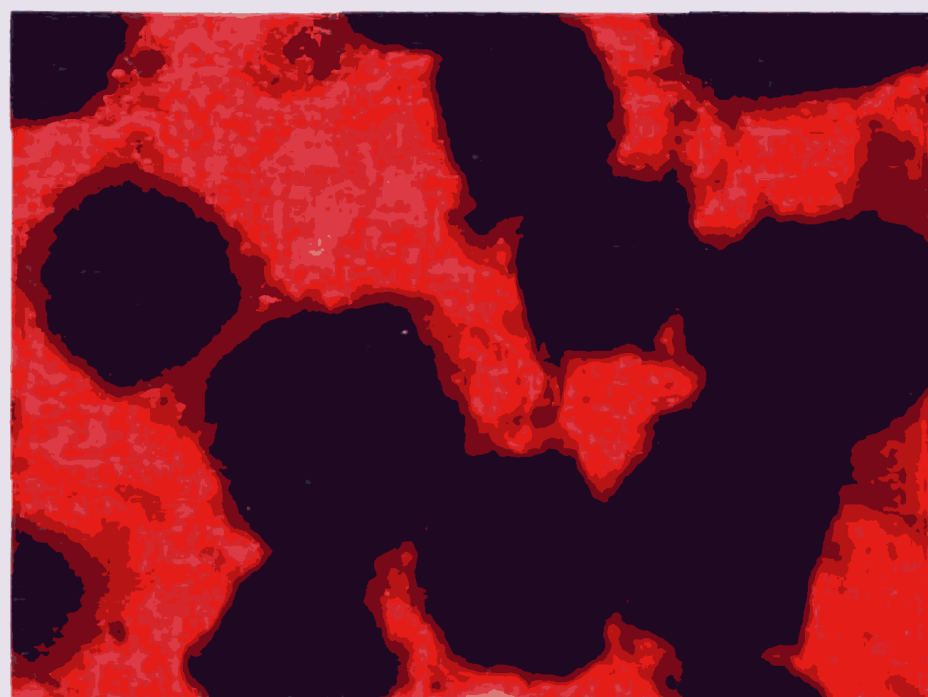
B



C



D



E

presence of sclerotia in the pith region and outer of the roots just below the bark region. The microscopic observations, denote that mycelial cells of hyphae are septate, branched and light to dark brown in colour. Sclerotia are brown to jet black and variable in shape i.e. round to oval and irregular.

4.4 Isolation, Purification and Identification of *Rhizoctonia bataticola*

Isolation from dry root rot affected plants was done on PDA medium. The isolates were identified on the basis of morphological characters, purified by using the hyphal tip technique and incubated at 25°C for further growth. The purified culture was maintained on PDA slants.

4.6 Pathogenicity test

Pathogenicity test was conducted by using susceptible variety BG 212 through pot inoculation technique. Plant showed clear cut symptoms of dry root rot. Primary and the secondary roots became black, distorted and brittle. On reisolation from the roots of infected plants exhibited the growth of *Rhizoctonia bataticola* on PDA.

4.7 Host resistance

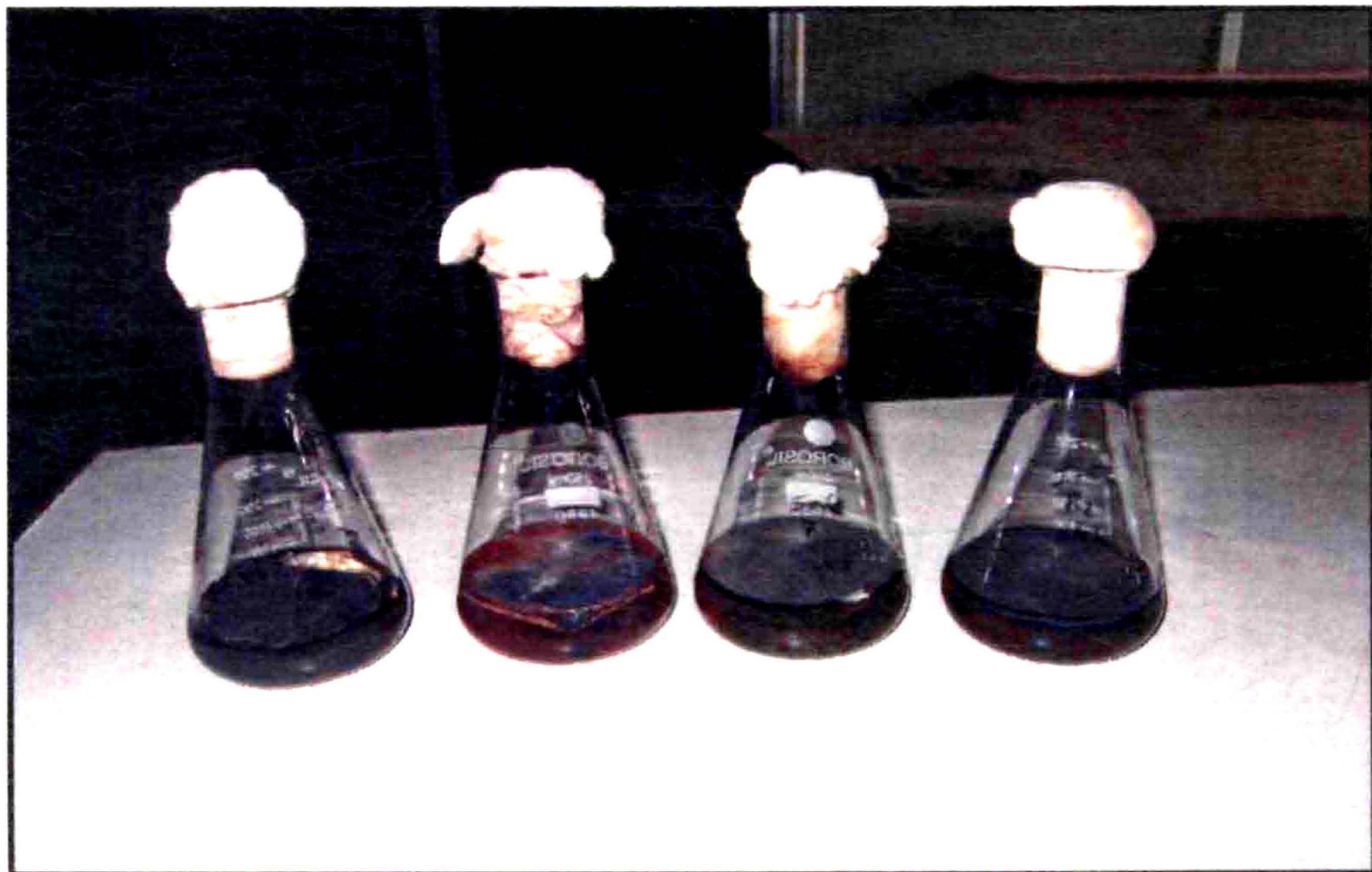
4.7.1 Screening of chickpea entries/genotypes against dry root rot

To locate the sources of host resistance against dry root rot pathogen 103 selected entries/genotypes were collected from AICRP on chickpea, JNKVV, Jabalpur, 29 from ICRISAT,

2. F) Growth of *Rhizoctonia bataticola* on Potato dextrose
broth

G) Trays showing seedlings of test line inoculated with
Rhizoctonia bataticola in an environmental test
chamber for 8 days at 35°C

Plate II



F



G

Hyderabad and 30 from NBPGR, New Delhi. These were screened alongwith susceptible check BG 212 through Blotter paper technique. The solution of mycelial mats of the pathogen was prepared in beakers for treating the seedlings. Seedlings inoculated with the culture of *R. bataticola* were kept into trays in an Environmental Test Chamber (ETC) for 8 days at 35°C which is depicted through Plate-2 : F and G.

The average mortality occurred in two replications is presented in the Tables 1 to 8. Disease reaction of isolate on chickpea entries/genotypes is shown in plate 3 and 4.

4.7.2 Entries/genotypes tested for dry root rot

The details of observations recorded for disease reaction on various genotypes collected from different sources are presented as below:

4.7.2.1 Donor

A set of 10 donor test lines of chickpea were evaluated against DRR (Table 1). The mortality ranged from 0-100 per cent. Only one entry BG 112 was found resistant to dry root rot, 7 of them were moderately resistant. BCP 17 and chaffa indicated tolerant and moderately susceptible reaction respectively.

4.7.2.2 Mutant

Screening of seven mutants of chickpea (Table 2) was done against DRR (*Rhizoctonia bataticola*). Data presented in the table clearly indicate that none of the mutant was resistant to dry root rot. All were moderately resistant except one (cracked) which was highly susceptible. The percent mortality ranged between 0 to 100.



4.7.2.3 ICRISAT Chickpea wilt and root rot nursery (CWRRN)

Evaluation of 29 genotypes of chickpea obtained from ICRISAT was done to record the incidence of dry root rot. The data furnished in Table 3 shows that 19 genotypes were noted to be resistant (<10 per cent mortality), 5 resistant to moderately resistant whereas ICC numbers 12233, 14409, 14436 and 11324, were moderately resistant and ICC 12450 was tolerant. The percent mortality ranged from 0 to 30.

4.7.2.4 National nursery-Dry root rot (DRRN)

Of the nineteen entries/genotypes tested only 6 viz., JG 74, CSJ 426, CSJ 427, CSJ 438, CSJ 440 and MPJG 99-115 were resistant, whereas BCP 21 and BCP 49 were highly susceptible. Seven of them were moderately resistant, one was tolerant and three were moderately susceptible. Mortality due to dry root rot was recorded between 0-100 per cent (Table 4).

4.7.2.5 National nursery wilt (NNW)

Data presented in Table 5 denotes that of the twenty genotypes collected from NNW ten were found to be resistant and eight resistant to moderately resistant against dry root rot whereas PG 00108 and IPC 2004-77 showed moderately resistant and tolerant behaviour. Mortality percentage varied from 0-30.

4.7.2.6 Station trial-Desi (S.T.D.)

Data pertaining to the disease reaction against dry root rot in the genotypes collected from station trial of AICRP on chickpea, JNKVV, Jabalpur is given in Table 6. It is evident from the data that 17 genotypes showed resistant reaction against the

disease, whereas eight were moderately resistant and ICCV 92944 and JG 2004-944 noted to be resistant to moderately resistant. The susceptibility of the disease ranged between 0 to 20 per cent.

4.7.2.7 Promising lines/Breeding material

Twenty promising entries of chickpea were evaluated. The observations recorded in Table 7 indicate that only five promising lines namely JG 9605, JG 210, S.T.B C₂ -25-1, JG 1-9 and C₂-25 were found to be resistant whereas 5 were highly susceptible. Seven entries of the rest were moderately resistant and three were tolerant. The mortality per cent ranged between 0 to 100.

4.7.2.8 Germplasm (NBPGR)

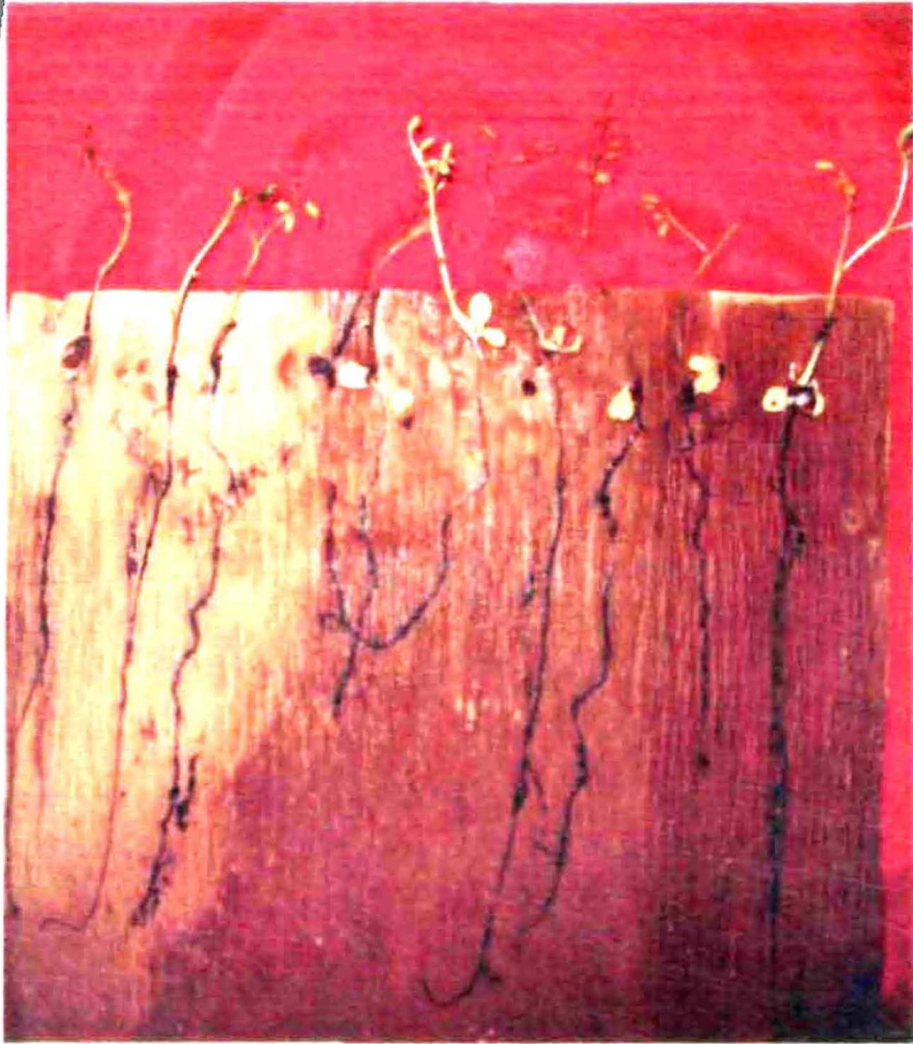
Thirty germplasm accessions procured from NBPGR, New Delhi were tested against dry root rot. The data presented in Table 8 clearly shows that of the 30 entries evaluated 9, 11 and 10 were found resistant, moderately resistant and tolerant, respectively. The susceptibility of the genotypes against dry root rot varied from 0 to 30 per cent.

The list of entries/genotypes noted to be resistant (66) among various entries/genotypes of chickpea evaluated for disease resistance against *Rhizoctonia bataticola* is given in Table 9.

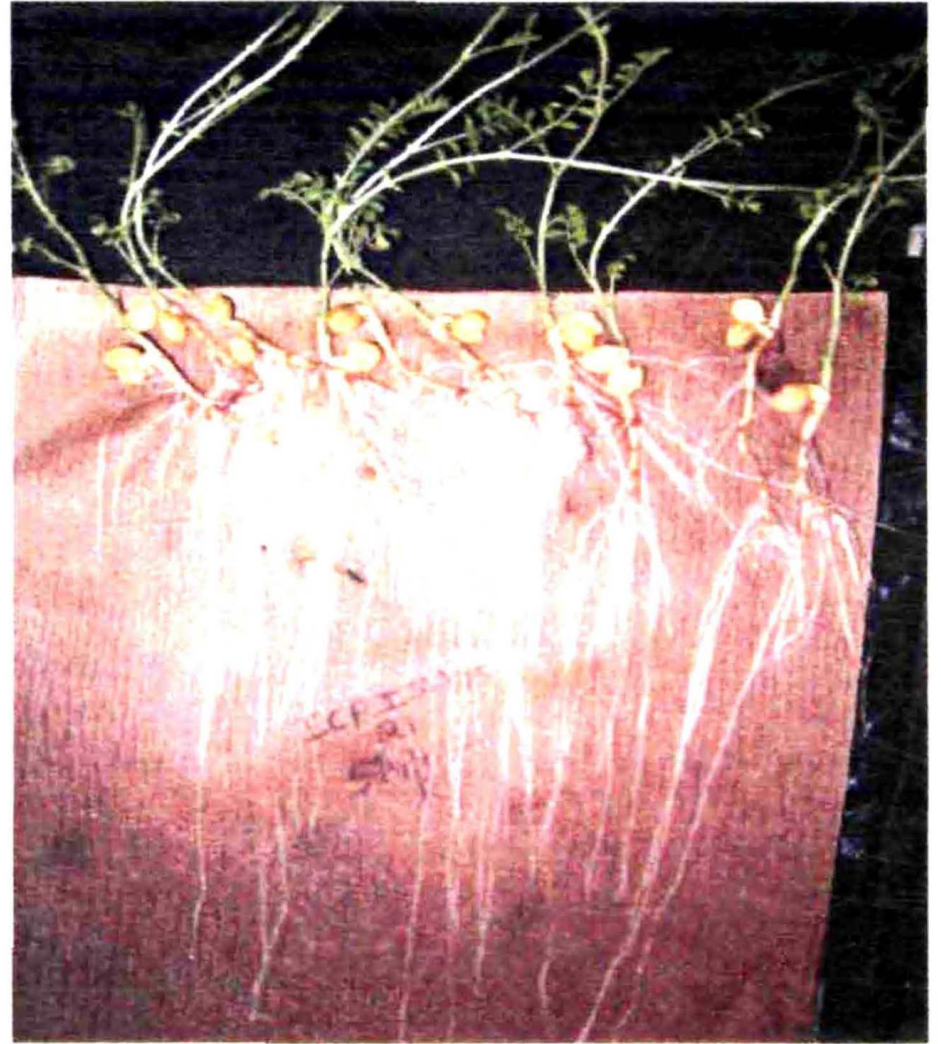
3. Disease reaction of isolate of *Rhizoctonia bataticola* on chickpea entries/genotypes alongwith susceptible check, BG 212 and control (rating scale : 1-9)

- | | |
|----|-------------------------------------|
| H) | Susceptible check : BG 212 |
| I) | Control : Uninoculated seedlings |
| J) | Resistant (R) : 1 |
| K) | Moderately Resistant (MR) : 2 and 3 |

Plate III



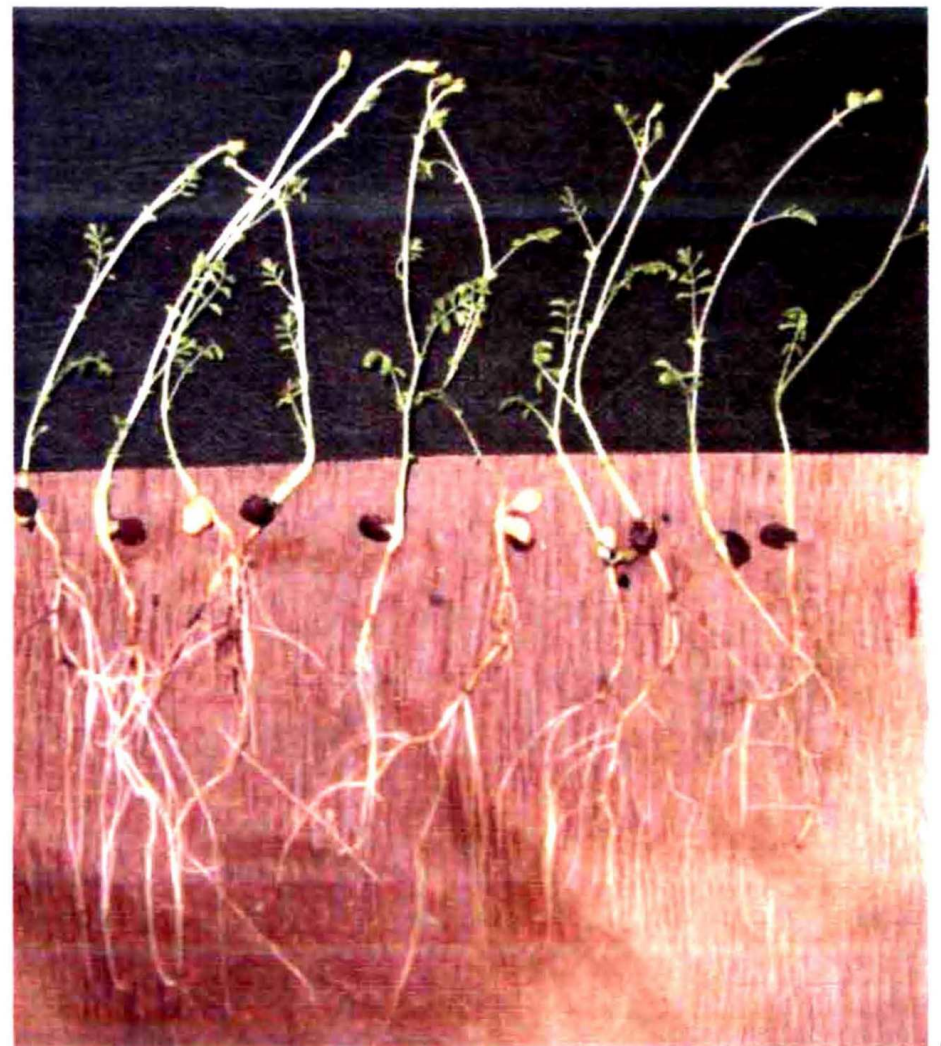
H



I



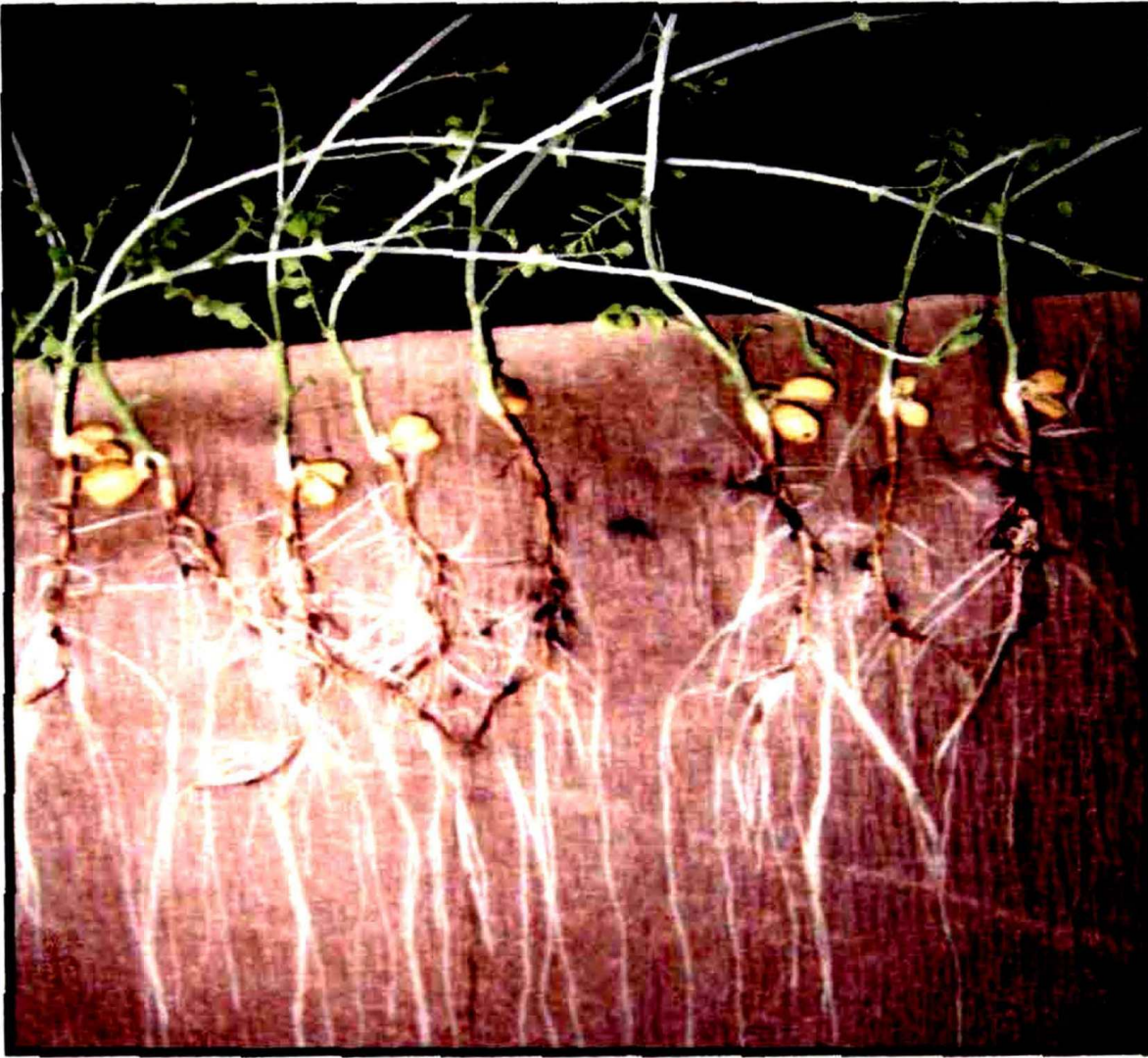
J



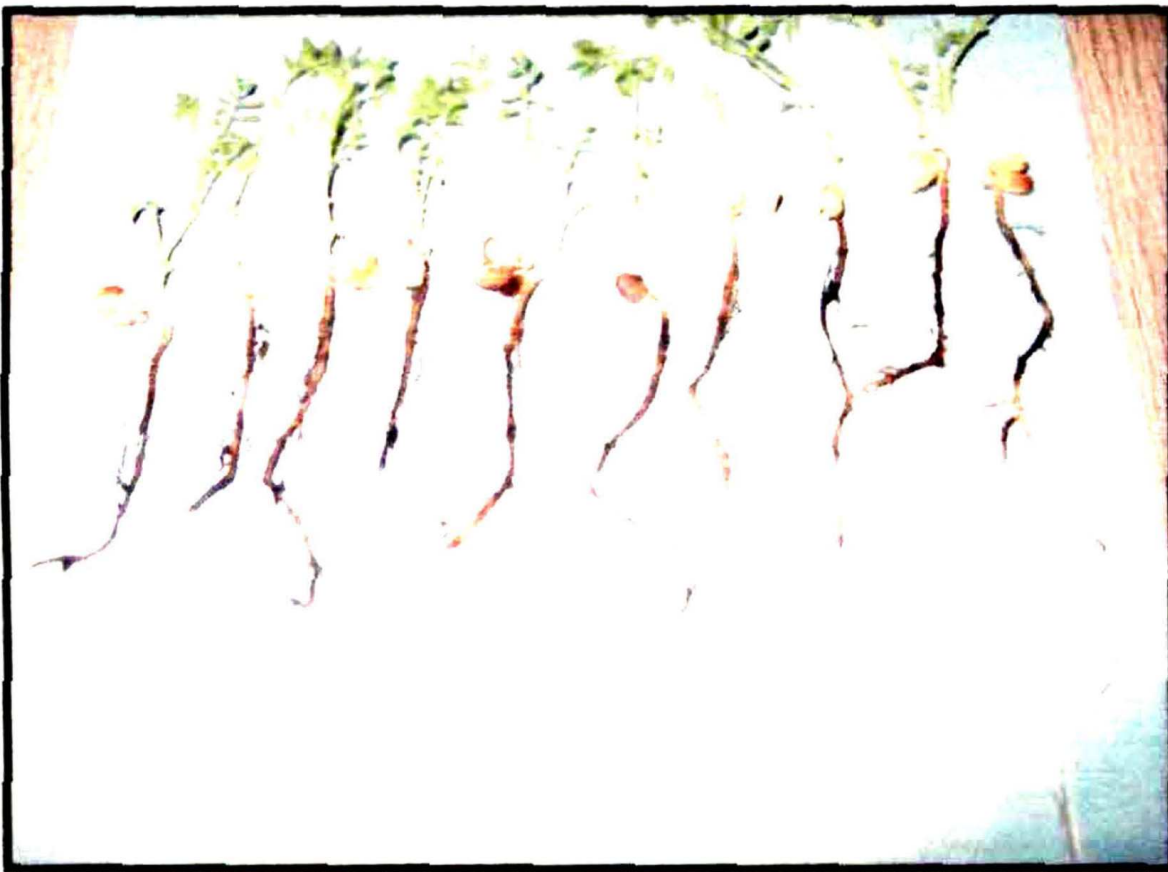
K

4. L) Tolerant (T) : 4 and 5
M) Moderately Susceptible (MS) : 6 and 7
N) Susceptible (S) : 8 and 9

Plate IV



L



M



N

Reaction of chickpea entries/genotypes against dry root rot through Blotter paper technique

Table 1. Donor : Entries-10

S. No.	Genotype	Rating	Category	*DRR%
1.	BG 112	1	Resistant	0
2.	BG 212	9	Susceptible	100
3.	BCP 17	5	Tolerant	5
4.	BCP 91	3	Moderately Resistant	0
5.	Chaffa	7	Moderately susceptible	35
6.	H 82-2	3	Moderately Resistant	0
7.	JG 74	3	Moderately Resistant	0
8.	MPJG 2002-108	3	Moderately Resistant	0
9.	MPJG 89-11551	3	Moderately Resistant	0
10.	MPJG 89-9023	3	Moderately Resistant	5

*Average of two replications

Table 2. Mutant : Entries-7

S. No.	Genotype	Rating	Category	*DRR%
1.	Simple leaf	3	Moderately Resistant	5
2.	Compacta	3	Moderately Resistant	0
3.	Broad few leaflets	3	Moderately Resistant	10
4.	A ₃ (MP x V ₁) x A ₆	3	Moderately Resistant	0
5.	Fasciated stem	3	Moderately Resistant	0
6.	Outward curved	3	Moderately Resistant	0
7.	Cracked	9	susceptible	100

*Average of two replications

Table 3. ICRISAT Chickpea Wilt, Root Rot Nursery (ICWRN): Entries-29

S. No.	Genotype	Rating	Category	*DRR%
1.	ICC 03104	1	Resistant	0
2.	ICC 03105	1	Resistant	0
3.	ICC 03106	1	Resistant	10
4.	ICC 03111**	1-3	Resistant to moderately resistant	15
5.	ICC 03112	1	Resistant	0
6.	ICC 03205	1	Resistant	0
7.	ICC 03206	1	Resistant	5
8.	ICC 03207	1	Resistant	10
9.	ICC 04107	1	Resistant	10
10.	ICC 04109	1	Resistant	5
11.	ICC 11322	1	Resistant	10
12.	ICC 11324	3	Moderately Resistant	20
13.	ICC 12233	3	Moderately Resistant	5
14.	ICC 12450	5	Tolerant	30
15.	ICC 12467	1	Resistant	0
16.	ICC 14344**	1-3	Resistant to moderately resistant	15
17.	ICC 14374	1	Resistant	0
18.	ICC 14376	1	Resistant	5
19.	ICC 14391	1	Resistant	10
20.	ICC 14404**	1-3	Resistant to moderately resistant	20
21.	ICC 14409	3	Moderately Resistant	15
22.	ICC 14432	1	Resistant	0
23.	ICC 14433	1	Resistant	0
24.	ICC 14434**	1-3	Resistant to moderately resistant	20
25.	ICC 14436	3	Moderately Resistant	5
26.	ICC 16124	1	Resistant	0
27.	ICC 950102-F ₄ -43P-BP**	1-3	Resistant to moderately resistant	20
28.	ICC 950106-F ₄ -49P-BP	1	Resistant	0
29.	ICC 950106-F ₄ -64P-BP	1	Resistant	10

*Average of two replications

**Resistant to moderately resistant

Table 4. National nursery dry root rot (DRRN) : Entries-19

S. No.	Genotype	Rating	Category	*DRR%
1.	BCP 21	9	Susceptible	100
2.	BCP 49	9	Susceptible	100
3.	BCP 114	3	Moderately Resistant	25
4.	CSJ 426	1	Resistant	0
5.	CSJ 427	1	Resistant	0
6.	CSJ 433	3	Moderately Resistant	35
7.	CSJ 437	3	Moderately Resistant	20
8.	CSJ 438	1	Resistant	0
9.	CSJ 440	1	Resistant	0
10.	CSJ 458	3	Moderately Resistant	30
11.	IPC 9391	3	Moderately Resistant	30
12.	IPC 98-22	5	Tolerant	5
13.	IPC 98-58	3	Moderately Resistant	0
14.	JG 74	1	Resistant	0
15.	MPJGK 3	3	Moderately Resistant	0
16.	PG 9425-5	7	Moderately susceptible	5
17.	PG 9425-9	7	Moderately susceptible	30
18.	MPJG 99-115	1	Resistant	0
19.	MPJG 2000-14	7	Moderately susceptible	45

*Average of two replications

Table 5. Promising entries from National nursery for wilt (NNW):Entries-20

S. No.	Genotype	Rating	Category	*DRR%
1.	GJG 0 205**	1-3	Resistant to moderately resistant	20
2.	GJG 0 312**	1-3	Resistant to moderately resistant	20
3.	GL 24080**	1-3	Resistant to moderately resistant	15
4.	GL 97016**	1-3	Resistant to moderately resistant	15
5.	IPC 2000-6	1	Resistant	10
6.	IPC 2004-54	1	Resistant	0
7.	IPC 2004-68	1	Resistant	0
8.	IPC 2004-77	5	Tolerant	25
9.	IPC 2005-28	1	Resistant	10
10.	IPC 2005-34**	13	Resistant to moderately resistant	20
11.	IPC 2005-36**	1-3	Resistant to moderately resistant	20
12.	IPC 2005-41**	1-3	Resistant to moderately resistant	15
13.	IPC 2005-59**	1-3	Resistant to moderately resistant	15
14.	IPC 2005-68	1	Resistant	0
15.	IPC-2005-74	1	Resistant	10
16.	MPJG 2001-04	1	Resistant	10
17.	MPJG 2004-03	1	Resistant	5
18.	PG 00108	3	Moderately Resistant	30
19.	PG 00109	1	Resistant	5
20.	Vishal	1	Resistant	10

*Average of two replications

**Resistant to moderately resistant

Table 6. Station trial-Desi : Entries-27

S. No.	Genotype	Rating	Category	*DRR%
1.	C ₂ 14-83	1	Resistant	0
2.	ICCV 87315	1	Resistant	0
3.	ICCV 87316	3	Moderately Resistant	10
4.	ICCV 92944**	1-3	Resistant to Moderately Resistant	20
5.	ICCV 95992	3	Moderately Resistant	20
6.	JG 63	3	Moderately Resistant	20
7.	JG 1-14	1	Resistant	5
8.	JG 1-18	3	Moderately Resistant	5
9.	JG 1-55	3	Moderately Resistant	0
10.	JG 2-125	1	Resistant	0
11.	JG 2-4-110	1	Resistant	5
12.	JG 2-14-1	3	Moderately Resistant	10
13.	JG 2-14-11	1	Resistant	5
14.	JG 2-14-115	3	Moderately Resistant	0
15.	JG 14-10	1	Resistant	0
16.	JG 2001-13	1	Resistant	0
17.	JG 2001-18	1	Resistant	0
18.	JG 2001-80	1	Resistant	5
19.	JG 2001-115	1	Resistant	5
20.	JG 2002-20	1	Resistant	5
21.	JG 2003-95	1	Resistant	0
22.	JG 2003-210	3	Moderately Resistant	15
23.	JG 2003-14-16	1	Resistant	0
24.	JG 2004-110	1	Resistant	5
25.	JG 2004-944**	1-3	Resistant to moderately resistant	20
26.	MPJG 99-115	1	Resistant	0
27.	MPJG 2003-14-2	1	Resistant	0

*Average of two replications

**Resistant to moderately resistant

Table 7. Promising lines/Breeding material : Entries-20

S. No.	Genotype	Rating	Category	*DRR%
1.	C ₂ 25	1	Resistant	10
2.	C ₂ 25 (IRC)	3	Moderately Resistant	0
3.	C ₂ 25-1	1	Resistant	0
4.	C ₂ 14-13-2	3	Moderately Resistant	0
5.	F ₆ -C ₂ -44	5	Tolerant	0
6.	F ₆ -C ₂ -97	9	Susceptible	100
7.	F ₅ MPJG 2003-95	5	Tolerant	0
8.	JG 210	1	Resistant	0
9.	JG 9605	1	Resistant	0
10.	JG 98003	3	Moderately Resistant	20
11.	JG 1-9	1	Resistant	0
12.	JG 2-25	3	Moderately Resistant	20
13.	JG 2000-7	9	Susceptible	100
14.	JG 2003-196	9	Susceptible	100
15.	JG 2003-14-16	5	Tolerant	10
16.	JG 2004-03	3	Moderately Resistant	0
17.	JG 2004-24	9	Susceptible	100
18.	JG 2004-110	3	Moderately Resistant	0
19.	MPJG 99-213	9	Susceptible	100
20.	MPx x IPC 9239 x JG 7	3	Moderately Resistant	0

*Average of two replications

Table 8. Germplasm NBPGR : Entries-30

S. No.	Genotype	Rating	Category	*DRR%
1.	IC 269240	3	Moderately Resistant	10
2.	IC 269263	1	Resistant	0
3.	IC 322-16	3	Moderately Resistant	0
4.	IC 327056	3	Moderately Resistant	0
5.	IC 327069	1	Resistant	0
6.	IC 327070	5	Tolerant	20
7.	IC 327073	1	Resistant	0
8.	IC 327074	5	Tolerant	30
9.	IC 327100	5	Tolerant	0
10.	IC 327104	5	Tolerant	20
11.	IC 327116	3	Moderately Resistant	5
12.	IC 327290	1	Resistant	0
13.	IC 327332	1	Resistant	0
14.	IC 327337	5	Tolerant	20
15.	IC 327364	5	Tolerant	20
16.	IC 327383	5	Tolerant	20
17.	IC 327393	5	Tolerant	0
18.	IC 327446	5	Tolerant	10
19.	IC 527582	5	Tolerant	15
20.	IC 327583	1	Resistant	0
21.	IC 327654	1	Resistant	10
22.	IC 327673	3	Moderately Resistant	10
23.	IC 327675	1	Resistant	0
24.	IC 327703	3	Moderately Resistant	0
25.	IC 327821	3	Moderately Resistant	0
26.	IC 327829	3	Moderately Resistant	0
27.	IC 327933	3	Moderately Resistant	10
28.	IC 327944	1	Resistant	0
29.	IC 327960	3	Moderately Resistant	0
30.	IC 327968	3	Moderately Resistant	20

*Average of two replications

Table 9. List of entries/genotypes scored as rating '1' against dry root rot <10 per cent

BG 112	CSJ 427	IC 327073	IC 327675
C ₂ 14-83	CSJ 438	IC 327290	IC 327944
C ₂ 25	CSJ 440	IC 327332	ICC 03104
C ₂ 25-1	IC 269263	IC 327583	ICC 03105
CSJ 426	IC 327069	IC 327654	ICC 03106
ICC 03112	ICC 04109	ICC 14391	IPC 2000-6
ICC 03205	ICC 11322	ICC 14432	IPC 2004-54
ICC 03206	ICC 12467	ICC 14433	IPC 2004-68
ICC 03207	ICC 14374	ICC 16124	IPC 2005-28
ICC 04107	ICC 14376	ICCV 87315	IPC 2005-68
IPC 2005-74	JG 2001-13	JG 2003-95	JG 2-14-110
JG 1-14	JG 2001-18	JG 2004-110	JG 74
JG 14-10	JG 2001-80	JG 210	JG 9605
JG 1-9	JG 2002-20	JG 2-125	MPJG 2001-04
JG 2001-115	JG 2003-14-16	JG 2-14-11	MPJG 2003-14-2
MPJG 2004-03	ICC 950106- F ₄ -49P-BP		
MPJG 99-115	ICC 950106- F ₄ -64P-BP		
PG 00109	Vishal		

Chapter VI

**Summary, Conclusion
and Suggestions**

DISCUSSION

Chickpea (*Cicer arietinum* L.) is an important protein rich legume crop and is cultivated as rainfed rabi crop in almost all the regions of the world. Besides, abiotic factors over 75 pathogens have been reported to cause diseases in chickpea (seed, soil and air borne) which are responsible for decreasing its productivity.

Among soil borne diseases dry root rot of chickpea caused by *Rhizoctonia bataticola* causes considerable losses (4-21 per cent) where the temperature exceeds $>25-30^{\circ}\text{C}$ during crop growth period particularly at podding and maturity stage. In the present study occurrence of the dry root rot of chickpea has been observed during flowering/podding stage of the crop.

According to a survey conducted to observe the incidence of dry root rot disease in chickpea cultivated area of Madhya Pradesh, it was observed that the disease appears at 2 phases of growth i.e. early seedling stage and flowering stage. Nema and Khare (1973) reported that chickpea cultivation in Madhya Pradesh was greatly threatened by soil borne diseases.

Gupta *et al.* (1983) reported the incidence of root rot ranging from 3.2 to 20.6 per cent in 30 villages of northern Madhya Pradesh. Nene *et al.* (1984) also recorded heavy losses to chickpea crop due to dry root rot caused by *R. bataticola*.

Studies on the diseased plants due to dry root rot exhibited straw-coloured appearance, forms reddish-brown sunken lesions on the stem, tap root and secondary roots becomes black near the soil line. Finally the roots of affected plants become brittle and broken easily. The pathogen was isolated from the infected plants identified, purified and conducted the pathogenicity test.

Present investigation was undertaken with a view to know the possible sources of host resistance against dry root rot of chickpea. Previously identified wilt resistant lines/genotypes alongwith other promising material and germplasm were evaluated through Blotter paper technique as described by Nene *et al.*, 1981. In the present investigation various entries/genotypes which comprises International chickpea wilt root nursery (CWRN), Dry root rot nursery (DRRN), germplasm National bureau of plant genetic resources, Station trials and promising entries from the AICRP on chickpea, JNKVV, Jabalpur were evaluated.

In all 162 entries/genotypes were evaluated and graded as resistant, moderately resistant, tolerant, moderately susceptible and susceptible (1-9 point rating scale) in which sixty six were scored as resistant against dry root rot. There was no symptom of any infection on these test lines as the root system was quite healthy, completely developed and exhibited white appearance and showed <10 per cent mortality. Further more, in 15 entries the disease rating is recorded 1 (resistant) but the average per cent mortality due to root rot has been observed in the range 10-20 per

cent. that's why they are scored as resistant to moderately resistant, (1-3 rating). Chickpea genotypes have been screened and identified as resistant to dry root rot by various workers on the basis of *in vitro* and field conditions (Nene *et al.*, 1981, Anand Rao and Haware, 1987, Gupta, 1987, Reddy *et al.*, 1990, Bhatt, 1993 Gupta, 1997, Khalid and Ilyas, 2000, Gupta *et al.*, 2003 Prajapati *et al.*, 2003 and Gupta and Babbar, 2006).

Many lesions on roots were seen in moderately susceptible genotypes though the new roots were generally free from these lesions. Whereas infected roots were completely discolored in highly susceptible genotype BG 212 (check). Pandey *et al.* (2004) evaluated 29 chickpea germplasm accessions for resistance against dry root rot caused by *R. bataticola* under *in vitro* conditions. He observed that 22 lines were either susceptible or highly susceptible for dry root rot. The percent mortality of plants ranged between 0-100.

Investigation on the occurrence of possible sources of host resistance in desi and kabuli chickpea types against *R. bataticola* under *in vitro* (Blotter paper technique) condition indicated that very limited entries/genotypes showed resistant reaction i.e. 10 per cent. For an effective varietal improvement programme, it is desirable to test the material in the multiple disease sick field and *in vitro* testing then use them as donors with multiple desirable traits such as resistant to diseases, agro-morphological and quality traits.



Chapter V

Discussion

SUMMARY, CONCLUSION AND SUGGESTIONS

An investigation entitled "**Dynamics of host resistance against *Rhizoctonia bataticola* (Taubenh.) Butler causing dry root rot in chickpea**" was carried out to identify the possible sources of host resistance for dry root rot under *in vitro* conditions. The disease was found to occur at flowering/podding stage of the plant. Diseased plants exhibited straw colour appearance with blackened roots which later on becomes brittle. Isolations were made from the affected plants collected from the experimental area, Seed Breeding Farm, JNKVV, Jabalpur. The Blotter paper technique was employed for screening the different entries/genotypes against dry root rot.

The observations recorded on disease incidence indicated that out of 162 entries/genotypes evaluated for resistance to dry root rot 81 were found to be resistant [66 were resistant (<10 per cent mortality), 15 were resistant to moderately resistant (>10.1 to 20 per cent mortality)] and 51 were moderately resistant while remaining 16, 4 and 10 were tolerant, moderately susceptible and highly susceptible respectively.

6.1 Conclusion

Dry root rot of chickpea caused by *Rhizoctonia bataticola* is becoming a serious problem as the ambient day temperature is exceeding >25-30°C during the flowering/pod formation stage. Investigations on the dynamics of host resistance against dry root rot led to conclusion that of the 162 entries/genotypes evaluated

66 entries/genotypes revealed resistant reaction (rating-1) with <10 percent mortality.

The present study explore the possibility of using the genotype as donors parent in resistance breeding programme to incorporate the resistance to dry root rot in high yielding and wilt resistant varieties of chickpea.

6.2 Suggestion for further work

The entries/genotypes which were found resistant against dry root rot (*Rhizoctonia bataticola*) under *in vitro* conditions need to be tested in natural field conditions in infested soil under multilocation evaluation test.



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